

Quantum Mechanics – Homework 1

Name: _____ Due Date: 2026.09.16

This homework covers Lectures 1–3. Show the main steps in every calculation and include units in numerical answers. Unless otherwise stated, you may use

$$\begin{aligned} h &= 6.626 \times 10^{-34} \text{ J s}, & \hbar &= 1.055 \times 10^{-34} \text{ J s}, \\ c &= 2.998 \times 10^8 \text{ m/s}, & k_B &= 1.381 \times 10^{-23} \text{ J/K}, \\ q_e &= 1.602 \times 10^{-19} \text{ C}, & m_e &= 9.109 \times 10^{-31} \text{ kg}. \end{aligned}$$

Problem 1. Measurements and Superposition (20 points)

The idealized color box has outcomes Black and White, and the hardness box has outcomes Hard and Soft. Each measurement is repeatable. Color and hardness are uncorrelated: selecting either outcome of one measurement gives a 50–50 distribution when the other property is measured.

- (a) An electron beam is prepared by selecting the White output of a color box. If color is measured again immediately, what probabilities are obtained for Black and White? Explain which experimental property you used. (3 points)
- (b) The White electrons are instead sent through a hardness box. The Soft output is selected and then sent into a final color box. Find the probabilities for Hard and Soft at the intermediate box and for Black and White at the final box. If 1200 White electrons enter the hardness box, how many are expected at each output of the final color box? (5 points)
- (c) Now replace the completed hardness measurement by the reversible two-path apparatus from Lecture 1. It separates the White beam into Hard and Soft paths and then recombines them before a final color measurement. State the observed Black and White probabilities when both paths are open. (3 points)
- (d) In the apparatus of part (c), the Soft path is blocked. Of 1200 incident White electrons, how many are expected to be absorbed, detected as Black, and detected as White? (4 points)
- (e) A classical mixture assumes that each electron secretly takes exactly one path, with

$$P(H) = P(S) = \frac{1}{2}, \quad P(B | H) = P(B | S) = \frac{1}{2}.$$

Use total probability to find $P(B)$. Compare it with part (c) and explain why the result requires a superposition rather than a classical mixture. (5 points)

Self-study requirement. Before answering Problem 2, study *Course Notes, Lecture 2*, subsection “Blackbody Radiation and Planck’s Hypothesis.” After completing the reading, answer the questions below.

Problem 2. Blackbody Radiation (20 points)

A blackbody may be modeled by a small hole in a heated cavity. The classical Rayleigh–Jeans spectral energy density and Planck’s spectral energy density per unit frequency are, respectively,

$$u_{\text{RJ}}(\nu, T) = \frac{8\pi\nu^2}{c^3} k_B T, \quad u_{\text{P}}(\nu, T) = \frac{8\pi h \nu^3}{c^3} \frac{1}{e^{h\nu/(k_B T)} - 1}.$$

- Explain why radiation emerging from a small hole in a cavity is a useful approximation to blackbody radiation. State two observed changes in the spectrum when the temperature is increased. (4 points)
- Let $x = h\nu/(k_B T)$. In the low-frequency limit $x \ll 1$, use $e^x - 1 \simeq x$ to show explicitly that Planck’s formula reduces to the Rayleigh–Jeans formula. (5 points)
- Describe the high-frequency limits of the two formulas. Explain why the Rayleigh–Jeans result leads to the ultraviolet catastrophe and how Planck’s formula avoids it. (5 points)
- Planck proposed that a mode of frequency ν exchanges energy only in amounts

$$E_n = nh\nu, \quad n = 0, 1, 2, \dots$$

Explain qualitatively why high-frequency modes are difficult to excite at a fixed temperature, while the classical equipartition assumption misses this suppression. (3 points)

- For the frequency form of Planck’s law, the maximum occurs at $x = h\nu_{\text{max}}/(k_B T) \simeq 2.821$. Calculate ν_{max} for $T = 5800 \text{ K}$. By what factor does ν_{max} change if the temperature is doubled? (3 points)

Problem 3. The Photoelectric Effect (20 points)

A metal has work function $W = 2.30 \text{ eV}$. It is illuminated by monochromatic light of frequency $\nu = 8.00 \times 10^{14} \text{ Hz}$. Use $h = 4.136 \times 10^{-15} \text{ eV s}$ for this problem. Einstein's photoelectric equation is

$$K_{\max} = h\nu - W = q_e|V_0|.$$

- (a) Calculate the threshold frequency ν_0 for this metal. (4 points)
- (b) Calculate the photon energy and the maximum kinetic energy of the emitted electrons, both in electron-volts. (4 points)
- (c) Find the magnitude of the stopping voltage $|V_0|$. Recall that an electron gains or loses 1 eV of energy when it moves through a potential difference of 1 V. (4 points)
- (d) The light intensity is doubled without changing its frequency. State how the photocurrent, $K_{\max}$, and $|V_0|$ change. Briefly justify each answer using the photon picture. (4 points)
- (e) The source is replaced by light of frequency $4.00 \times 10^{14} \text{ Hz}$, but its intensity can be made arbitrarily large. Will electrons be emitted? Explain quantitatively. (4 points)

Problem 4. Young’s Double-Slit Experiment (20 points)

Monochromatic light of wavelength $\lambda = 500$ nm passes through two narrow slits separated by $\ell = 0.250$ mm. A screen is placed $D = 2.00$ m from the slits.

- (a) Starting from the path difference $\Delta r = \ell \sin \theta$, derive the condition for constructive interference. Then use the small-angle approximations $\sin \theta \simeq \tan \theta \simeq y/D$ to show that

$$y_n \simeq \frac{D\lambda}{\ell} n, \quad n \in \mathbb{Z}. \quad (5 \text{ points})$$

- (b) Calculate the positions of the $n = 0, \pm 1, \pm 2$ bright fringes and the spacing Δy between neighboring bright fringes. (4 points)

- (c) If the two slit amplitudes are $A_1 = A_0 e^{i\theta_1}$ and $A_2 = A_0 e^{i\theta_2}$, expand $|A_1 + A_2|^2$ and show that

$$I(y) \propto 2A_0^2 [1 + \cos(\theta_1 - \theta_2)].$$

Identify the term responsible for interference. (5 points)

- (d) Evaluate the intensity at the center $y = 0$ and halfway between the central maximum and the first bright fringe, using

$$I(y) = 2I_0 \left[1 + \cos \left(\frac{2\pi \ell y}{\lambda D} \right) \right].$$

Identify each point as bright or dark. (3 points)

- (e) Electrons are now sent through the slits one at a time. Describe what is observed in one detection and what appears after many detections. What happens to the interference pattern if an apparatus leaves a reliable record of which slit each electron used? (3 points)

Problem 5. Deriving the de Broglie Relation (20 points)

The Planck–Einstein relation for a photon is $E = h\nu$. Light also satisfies $E = pc$ and $c = \lambda\nu$.

- (a) Combine the three relations above to derive

$$p = \frac{h}{\lambda}.$$

Then use $k = 2\pi/\lambda$ and $\hbar = h/(2\pi)$ to show that $p = \hbar k$. (5 points)

- (b) State de Broglie’s hypothesis for matter. Carefully distinguish the part derived for photons in part (a) from the new physical hypothesis made for a massive particle. (3 points)

- (c) An electron starts from rest and is accelerated through a potential difference $V_{\text{acc}} = 150 \text{ V}$. Starting from

$$q_e V_{\text{acc}} = \frac{p^2}{2m_e},$$

derive

$$\lambda = \frac{h}{\sqrt{2m_e q_e V_{\text{acc}}}},$$

and calculate the electron’s momentum and de Broglie wavelength. (6 points)

- (d) Calculate the wave number k for the electron in part (c). Write a one-dimensional plane wave for an electron with this momentum traveling in the positive x direction, leaving the amplitude as A and the angular frequency as ω . (3 points)

- (e) If the accelerating voltage is increased from V_{acc} to $4V_{\text{acc}}$, by what factors do p , λ , and k change? Explain how electron diffraction provides experimental evidence for de Broglie’s hypothesis. (3 points)

End of Homework 1